

INFOSCI 103 | Nemuulen Togtbaatar

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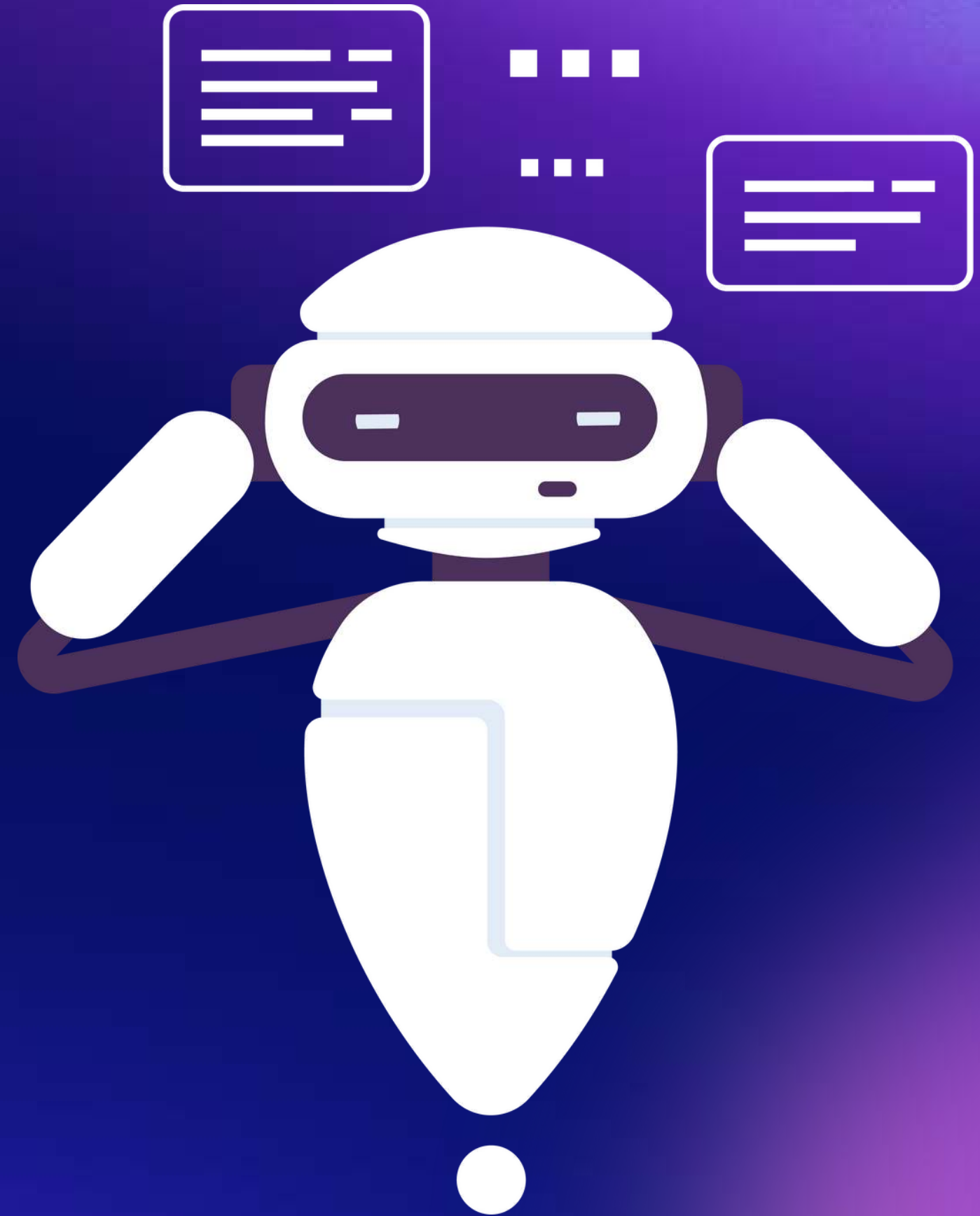
“PHANTOM FRIEND” OR “JUST A BOX WITH INFORMATION”

Personification and Ontological Categorization
of Smart Speaker-based Voice Assistants by
Older Adults (Pradhan et al, 2019)



Starter Questions

1. Do you personify AI? How and why do you do that? How would you assign a gender?
2. Do you say 'please' or 'thank you' when chatting with AI or voice assistants? Do you feel any difference in response?





Goals of study:

- 1) HOW do older adults personify voice assistants in their homes?
- 2) WHEN do older adults categorize voice assistants as “object-like” vs. “human-like”
- 3) WHAT are older adults’ perceptions of voice assistants in terms of filling social roles?

Contributions of this paper:

- 1) an understanding of how participants in this study ontologically categorize voice assistants
- 2) recommendations for designing with ontological category in mind for older adults
- 3) a discussion of the personification of voice assistants in relation to previous established theories of anthropomorphism and ontological categorization.

Background research

Anthropomorphism of Technology

attribution of human traits, emotions, or intentions to non-human entities.

✓ therapeutic benefits

✗ reduced human interaction

✗ danger of manipulation

✗ reveal personal information

★ Politeness towards devices is concerned as “overlearned social behaviors” that are not due to “conscious beliefs that computers are human or human-like”. (Nass and Moon, 2000)

★ Voice or spoken dialogue systems, and interfaces that have a “face” and physical space support more anthropomorphism

★ Many people mentioned Alexa as “she”, “her”, “bff”, “new best friend”, and “someone to talk to” and were regularly greeting.

★ Users in multi-user households are more likely to personify them.

Methodology

Placed Amazon Echo Dot Devices in the homes of **seven** individuals aged **65** or older for **three weeks**.



Overview of participants:

- Over 65 years old (65–83)
- No prior experience
- Low technology use (less than once a day)
- All participants lived alone, except for P6, who lived with her son.
- Five participants lived in senior living communities
- Three of these participants lived in low income housing
- Two participants lived in their own homes.

ID	Age	Gender	Education	Quality of life overall	OPQOL score
P1	65	Male	High school	Good	49
P2	75	Female	High school	Alright	58
P3	71	Female	Some college, no degree	Alright	56
P4	65	Female	Some college, no degree	Good	52
P5	71	Female	High school	Very good	56
P6	72	Female	Less than high school diploma	Good	52
P7	83	Female	High school	Very good	65

Table 1. Participants' demographics

Methodology & Limitations

- Weekly in-person interviews
- Daily diary check-ins via phone calls
- Collected usage logs

Used constructivist grounded theory approach to analyze interviews. They looked for key behaviours such as “attributed as friend”, “somebody in house”, “use of pronouns”, and politeness.



- ✓ Forced adoption allowed to see changes over time
- ✗ Daily calls may reminded them to use the device more frequently
- ✗ Did not intend to obtain generalizable results - few participants

Results

Main findings:

A. Personification and Objectification of Voice Assistant

1. Inconsistent Use of Personal Pronouns.
2. Polite Behaviors and Personification.
3. Neither human nor object.

B. The Social Role of Alexa: Movement between Ontological Categories

1. Qualities of Interaction: Natural Language and Responsiveness.
2. Qualities of the User: Perceiving a Personalized Approach and Desire for Companionship.
3. Changes over the Longer Term, Location and from Moment to Moment.

Analyzed using “three factor theory of anthropomorphism”



1

Inconsistent Use of Personal Pronouns

- All participants in the study referred to Alexa as “she” or “her” multiple times during interviews.
- Used “she” and “it” in the same sentence -> users might **not binarily classify voice assistants**

“Sometimes in the morning, after I say, ‘Good morning,’ it might say something that makes me laugh, sometimes, I will call her name and say, ‘Oh that was so funny, it really made me laugh’ and it doesn't respond”. (P2, week 2-end, interview)



Polite Behaviors and Personification

- Found that most participants in our study – all but P3 and P6 – used polite terms, such as “please” and “thank you” when using Alexa.
- It’s learned over time and mostly “**social mindless responses**”

“I guess it's just something that comes naturally to me or something, and I was always raised by my mother to say, thank you. So, I guess it's something instilled in me”. (P4, week 3-end, interview)

3

Neither human nor object

- People see robots as “in-between” human and object
- Shifting between these categorizations

For P2, Alexa was, “...like a **companion**, that's not there. So, you have to make it... not a fake, but a **phantom friend**” (P2, week1-end, interview).

P6 further described how “**talking**” to Alexa made her feel as if she were talking to “**somebody**”, but she also acknowledging that Alexa was **not real**.

P5 refused to categorize Alexa at all, instead designating a new category: “I think of it as that’s **Alexa**”.

1

Qualities of Interaction: Natural Language and Responsiveness

- Natural language conversation and the **responsiveness of these devices is what make them social.**
- Alexa lacked responsiveness at times -> object-like.
- Intentional choice of designers to assign a particular name and voice to the assistant -> triggering personification.

"...when it talks, I don't see a box. I just see... It's like somebody is standing there talking to me... It's making my day. I say, good morning. And they say, good morning [name], how are you today?" (P2, week 3-end, interview).

"It do more asking me questions than listening and responding. That's really not chat-chat companion. It's just asking what I like. It's a good object to have. (P1, week 3-end, interview)

2

Qualities of the User: Perceiving a Personalized Approach and Desire for Companionship

- Alexa addressed them by using their name which they felt was “personal”.
- Conversations are sometimes meaningful
- Individuals may be more likely to personify agents when they feel lonely.
- **Filling a social need.**

“And it calls my name, “Good morning [P2]”. Oh! I like that. It means so much... Because, when somebody is talking to me, call my name. Don't just talk to me, acknowledge me”. (P2, week 1-end, interview)

P3 said that she felt that she, “wasn't in the house alone”, because “I had somebody I could talk to” (P3, week 1-end).

“When you live alone and after you've been married for a long time, you miss the little things like that sometimes” (P4)

3

Changes over the Longer Term, Location and from Moment to Moment

- Participants' perceptions of where Alexa fit in terms of ontological **categorization shifted over time.**
- The **physical location** of Alexa also affects these perceptions.



P7 went from referring to Alexa as “company”, in the first week to describing it as talking to a program something” at the exit interview.

P3, who initially felt “odd talking to a machine”, with time became comfortable such that she considered Alexa “an invisible friend, somebody I can talk to one on one”.

Discussion

1. Understanding of the relationship between anthropomorphism, embodiment and voice assistants

- + understand how other user groups in general perceive these voice assistants
- + analyze diverse life contexts, including cultural, geographical, philosophical, political, and religious perspectives

2. People's perceptions going forward are affected by the device's design

- + study whether harmful values are being perpetuated through these designs

3. Understanding of intricate changes in human behaviors for virtual agents

- + examine whether a fluid ontological categorization takes place with technologies aside from voice assistants, such as robots.
- + animate projections such as a holographic projection or more anthropomorphic forms when using

Discussion

4. Designers “purposefully design the experience they want users to perceive” through the creation of personas

- + not all participants desire the female voice.
- + designers must consider the ways individuals’ dynamics and the roles

5. Physical location of the device and its embeddedness in the home environment matters. Device personification is linked to “always on” features

- + Examining other sensory modes of feedback might enhance use of the device.
- + important to consider home as a sacred space

My suggestions: (How this study could be useful?)

- + Can assist on studying on how voice assistants help older people with dementia or Alzheimer’s. (new feature)
- + Devices having sleepy face when off and wake up when they get activated

Discussion

- 👍 Really focused and gave attention to every detail
- 👍 The purpose of research is to make technology accessible to older adults by providing suggestions based on findings
- 👍 Highly influential, cited by 200 papers.
- 👎 Unclear methodology
- 👎 Cannot understand all different types of older adults (e.g. with more tech use, lives with big families)
- 👎 6 out of 7 participants were female
- 👎 Trial usage of Alexa - like an ad from Amazon - possible ethical problem



Summary:

- Older adults mostly classified voice assistants **in-between** human and machine (analyzed based on personal pronouns, politeness, responsiveness)
- **Natural language use** and **interface design** was triggering personification.
- **Location** matters to people's perception of voice assistants.
- It sometimes fills **social or emotional needs**
- Participants' perceptions of where Alexa fit in terms of ontological **categorization** **shifted over time.**

Questions:

1. If they had face, would you personify them more? Why and why not?
2. How could voice assistants develop in the future? What would be the positive and negative effects?
3. Do you think AI or humanoid robots can replace human partners in the future if they can fill emotional needs?



Thank You!



Reference:

Alisha Pradhan, Leah Findlater, and Amanda Lazar. 2019. "Phantom Friend" or "Just a Box with Information": Personification and Ontological Categorization of Smart Speaker-based Voice Assistants by Older Adults. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (Nov. 2019), 214:1–214:21. <https://doi.org/10.1145/3359316>